

CSS/CSX

General information on commissioning, integration and operation of devices in IO-Link mode



Described product

IO Link - CSS/CSX

Manufacturer

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Original document

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1 About this document

1.1 Purpose of this document

The ISDU descriptions in this document apply to IO-Link-capable sensors.

In some cases, functions may be described in this document which are not supported by individual sensors. The functions in question are marked accordingly (see "Symbols", page 4).

The individual range of functions of a single sensor is shown in full in the IODD overview on the respective product page at www.sick.com.

1.2 Intended use

Use IO-Link only as described in this documentation.

1.3 Symbols



NOTICE

This symbol indicates important information.



NOTE

This symbol provides additional information, e.g., dependencies / interactions between the described function and other functions, or when individual functions are not supported by every sensor.

2 Description of IO-Link

IO-Link communication interface

The product has the IO-Link communication interface.

IO-Link communication is a **Master-Device** communication system.

The sensor can be used in standard I/O mode (SIO) or in IO-Link mode (IOL). All automation functions and other parameter settings are effective in IO-Link mode and in standard I/O mode.

The following functions are supported via this standard IO-Link communication interface:

- Flexible sensor settings
- Digital transmission of the sensor signals to the **IO-Link Master**
- Visualization and configuration of the sensor
- Diagnosis / **Condition Monitoring**
- Device identification
- Easy device replacement
- **Events**

A detailed description of the adjustable functions and associated indices can be found in the technical information "IO-Link description", available for download at www.sick.com.

IO-Link and control integration

IO-Link is a non-proprietary internationally standardized communication technology which makes it possible to communicate with sensors and actuators in industrial environments (IEC 61131-9).

IO-Link devices communicate with higher-level control systems via an **IO-Link Master**. The IO-Link devices are connected to these via a point-to-point connection.

IO-Link Master are available in different versions. In most cases, they are remote fieldbus gateways or input cards for the backplane bus of the control used.

For an **IO-Link Device** to communicate with the control system, both the **IO-Link Master** and the **IO-Link Device** must be created (integrated) in the hardware configuration in the control system manufacturer's engineering tool.

Not all manufacturers of control systems support the use of the IO-Link device description file (**IO-link Device Description = IODDs**). If a third-party **IO-Link Master** is used, the **IO-Link Device** can also be integrated by manually entering the relevant sensor parameters directly during hardware configuration.

To ensure that the **IO-Link Device** can be easily integrated into the control program, SICK also provides function blocks for many control systems. Among other things, these function blocks make it easier to read and write the individual sensor parameters and support the interpretation of the process data supplied by the **IO-Link Device**. You can also download them free of charge from the homepage: www.sick.com.

A number of tutorial videos are available on SICK's YouTube channel to assist with the integration of SICK **IO-Link Master**: www.youtube.com/SICKSensors.

If you have any questions, SICK Technical Support is available to help all over the world.

3 Documentation and accessories

Accessory components and additional information are available for integrating and setting the IO-Link device. You will find documentation and software, accessories and links to the **SICK Product ID**.

SICK product ID

The SICK product ID uniquely identifies the product. It also serves as the address of the web page with information on the product.

The SICK product ID comprises the host name pid.sick.com, the part number (P/N), and the serial number (S/N), each separated by a forward slash.

For many products, the SICK product ID is displayed as text and QR code on the type label and/or on the packaging.



Figure 1: SICK product ID

Documentation and software

- IODD: Device description file
- IODD overview: List of IODD contents
- IO-Link description: Detailed description of the process, service data and events of the IO-Link device
- SOPAS ET: Configuration software as a free download
- The documentation for SOPAS ET is stored in the system folder on your computer with the download:
C:\Program Files (x86)\SOPAS ET\help
- Visualization file (SDD = **SOPAS Device Description**) for operation via SOPAS ET.
- **Function Block Factory**

IO-Link products can be easily connected to a computer via USB using the **SiLink master**. You can quickly and easily test or parameterize the connected products using the **SOPAS ET (SICK Engineering Tool** with graphic user navigation and convenient visualization).

Accessories

- IO-Link master
- SiLink2 Master, www.sick.com/1061790
- Connecting cables

4 Physical layer

The device data is automatically communicated to the **IO-Link Master** . It is important to ensure that the **IO-Link Master** used supports this performance data.



NOTICE

The maximum current consumption of the **IO-Link Device** (including load at the outputs) must not exceed the permissible output current of the respective port on the **IO-Link Master** .

Table 1: Physical layer – System data

SIO Mode	Yes
Min. Cycle Time	1.8 ms
Baud rate	COM 3 (230.4 kbit/s)
Process Data Length PD In (from Device to Master)	12 bytes
IODD Version	V1.0
Supported IO-Link Version	1.1.0

5 Integration of the sensor into the control level

Connecting the IO-Link device to the IO-Link Master

To operate the product in IO-Link mode, it must be connected to a suitable **IO-Link Master**. This is used for further integration into the control system.



NOTE

The cable length between the **IO-Link Master** and **IO-Link device**: maximum 20 m.

To enable the **IO-Link device** to be taken into account in the automation structure, the device must be registered by the **IO-Link Master** or PLC manufacturer via an **engineering tool** and the communication relationship must be parameterized. The device description IODD can be used for this purpose. It contains information on identification, device parameters, process and diagnostics data, communication properties, and more.

Download the **IODD-File** from www.sick.com or from the **IODD-Finder** of the IO-Link consortium (**IODD finder**). Make sure you always use the latest **IODD-File**.



NOTE

After successful connection of the product to the **IO-Link Master**, the green (Power) LED flashes to indicate a functioning IO-Link communication between the **master** and **device**.

6 Accessories for visualization, parameterization and integration

IO-Link products can be easily connected to a computer via USB using the **SiLink Master**. The connected products can be tested or parameterized quickly and easily using the program **SOPAS ET** (**SICK Engineering Tool** with graphical user guidance and convenient visualization).

The following IO-Link-specific information is available at www.sick.com:

- SOPAS ET as a free download
- Visualization file (SDD = **SOPAS Device Description**) for operation via SOPAS ET.
- Overview of available **IO-Link Master** for the integration of **IO-Link Devices**.

7 Setting, configuration and integration

In addition to the manual setting on the device, the sensor can also be configured via IO-Link.

Configuration via the REST interface of the **IO-Link Masters** is also possible. Please read the operating instructions for the **IO-Link Masters**.

A list of all functions that can be configured can be found in the IODD and the IO link overview.

Setting options

Setting via buttons (limited setting options if necessary)

Configuration via IO-Link

1. Setting via **SiLink2 Master** (SOPAS ET)
2. Setting via **IO-Link Master** (PLC)
 - **IO-Link Master** from the PLC manufacturer
 - **IO-Link Master** from third-party manufacturer (SICK), more manual effort

Integrating the IO-Link device into the PLC

To simplify programming in the PLC, device-specific function blocks can be generated via the **Function Block Factory**.

Function blocks simplify acyclical communication (service data communication) between the PLC and **IO-Link Device** and the interpretation of process data. They provide device parameters and correct device data types and translate the parameters provided into indices and sub-indices.

8 Process data structure

Download the **IODD-File** from www.sick.com or from the **IODD-Finder** of the IO-Link consortium (**IODD finder**). Make sure you always use the latest **IODD-File**.

Process data is transmitted cyclically. There is no confirmation of receipt.

The cycle time is set by the master, whereby the minimum cycle time of the sensor cannot be undercut.



NOTE

The service data (acyclic data) does not influence the cycle time.

Process data structure

Depending on the application requirements, the process data content can be changed via parameter index 120 (Process Data Select), [see table 14](#).

Evaluation Mode (Default), Color Match Value (CMV) + Q1 ... Q4 + all Qint status

Index 120 = 0

Table 2: Process data structure - Evaluation Mode

Byte number	0	1	2	3	4	5	6	7	8	9	10	11
Function Slot	CMV QL4		CMV QL3		CMV QL2		CMV QL1		Qint status		QL + diagnostics	
Data type	Uint16		Uint16		Uint16		Uint16					
CMV area	0...999		0...999		0...999		0...999		Bitmap see table 4, page 12			

The CMV value shows the degree of match to the taught-in color. The sensitivity setting (switching threshold) refers to the CMV value:

999 = full match

0 = no match

Measurement Mode, Lab / RGB measurement + Q1 ... Q4 + all Qint status

Index 120 = 1

2 options for the measurement mode when using ISDU 296:

- RGB = Receive signal of the red-green-blue channel
Index 296 = 0 (default)
- Lab = standardized Lab color space, based on L = lightness, a = green/red scale, b = blue/yellow scale
Index 296 = 1 (values in 1/100)

For interpretation of the data, see [table 3](#) RGB: black: 0 - 0 - 0 / white: 255 - 255 - 255 / gloss: > 255

Lab color space: black: 0 - 0 - 0 / white: 100 - 0 - 0 / gloss: L-value > 100

The gloss level depends on the alignment angle of the sensor.

Table 3: Process data structure - Measurement Mode

Byte number	0	1	2	3	4	5	6	7	8	9	10	11
Function Slot	Red / L		Green / a		Blue / b		CMV QL1		Qint status		QL + diagnostics	
Data type	INT16		INT16		INT16		Uint16					
RGB range	0...255 ¹⁾		0...255 ¹⁾				0...999		Bitmap table 4 and table 5			
Lab range in 1/100	0...160.00/100		-128.00...+127.00/100									

¹⁾ With gloss or mirroring, the value rises above 255 up to the maximum value of the data type.

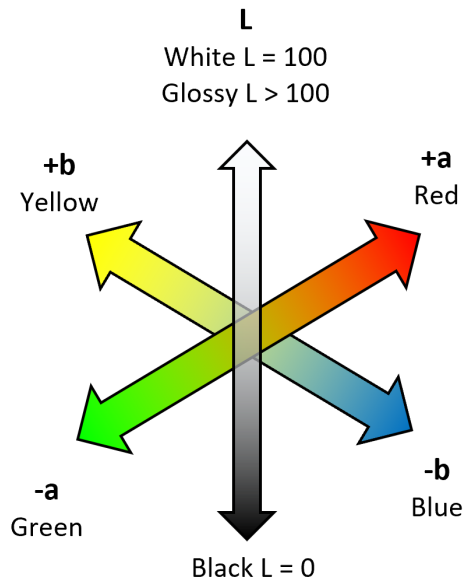


Figure 2: Lab color space

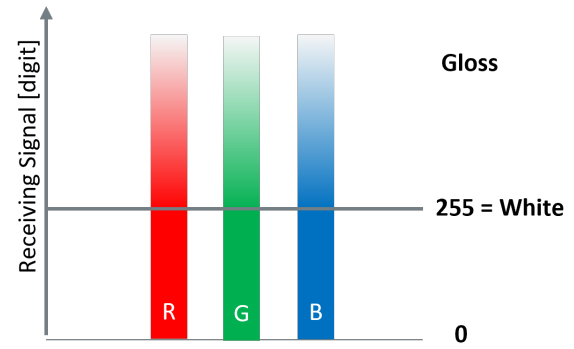


Figure 3: RGB color space

Table 4: Status - Qint

Byte number	Byte 8								Byte 9							
Bit offset	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bitmap	Qint 24	Qint 23	Qint 22	Qint 21	Qint 20	Qint 19	Qint 18	Qint 17	Qint 16	Qint 15	Qint 14	Qint 13	Qint 12	Qint 11	Qint 10	Qint 9
States	0 = False/OFF 1 = True/ON															

Table 5: Status - QL + Diagnostics + Qint

Byte number	Byte 10								Byte 11							
Bit offset	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bitmap	Qint 8	Qint 7	Qint 6	Qint 5	Qint 4	Qint 3	Qint 2	Qint 1	Reserved	Reserved	PD invalid	QoR Alarm	QL4	QL3	QL2	QL1
States	0 = False/OFF 1 = True/ON															

QL output: physically available on connector pins (number of physical QLs depending on model)

Qint output: only available via IO-Link process data

9 Service data

Service data is only exchanged between the controller and the IO-Link device via the IO-Link master on request from the controller (acyclical). The service data is referred to as ISDUs. With ISDU, users can read information about the status of the connected IO-Link device and/or write new parameters to change the configuration.

The corresponding counterpart confirms receipt of the data.

If the IO-Link device does not respond within five seconds, the master reports a communication error.

9.1 Device identification

System

Table 6: Device identification

ISDU			Name	Data type	Data repository	Length	Access	Default value
Index		Sub-Index						
DEC	HEX							
16	0x10	-	Vendor Name	String	-	7 bytes	ro	SICK AG
17	0x11		Vendor text			64 bytes		www.sick.com
18	0x12		Product Name			18 bytes		CSS-XXXXXX
19	0x13		Product ID			9 bytes		see ISDU 219
219	0xDB	0	Product ID	Record	-	7 bytes	ro	
		1	Product ID IO-Link Device	String		7 bytes		

Vendor Name Manufacturer name as UTF-8 string

Vendor text Manufacturer text as UTF-8 string

Product Name Complete type designation of the connected sensor as UTF-8 string

Product ID The **Product ID** is also the order number of the connected IO-Link device. The **Product ID** can be found under ISDU 219 in order to provide a family IODD for a device family.

Table 7: Device identification - Product Text / Serial Number

ISDU			Name	Data type	Data repository	Length	Access	Default value
Index		Sub-Index						
DEC	HEX							
20	0x14	-	Product Text	String	-	64 bytes	ro	Color Sensor
21	0x15		Serial number			8 bytes		YYWWnnnn

Format of the serial number:

YYWWnnnn (Y = year, W = week, n = sequential numbering)

Version

Table 8: Device identification - Version

ISDU			Name	Data type	Data repository	Length	Access	Default value
Index		Sub-Index						
DEC	HEX							
22	0x16	-	Hardware version	String	-	12 bytes	ro	Xnnn
23	0x17	-	Firmware version	String	-	12 bytes	ro	n.n.n.nnn.X.

These ISDUs display hardware and software versions.

n = consecutive numbering

Hardware version: X = letter with 3-digit number (UTF-8 string)

Software version: 5-digit with extension identifier, with dot separation (UTF-8 string)

User-specific entries

Table 9: Device identification - *Specific tag*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
24	0x18	-	Application-specific tag	String	yes	32 bytes	rw	*****	
64	0x40	-	Device-specific tag		no			*****	

You can save any text with a maximum length of 32 characters at **Application-specific tag**. This can be useful for describing the exact position or task of the sensor in the overall machine. The **Application-specific tag** is saved via the **Data repository**.

You can also save any text with a maximum length of 32 characters at **Device-specific tag**. This indicator is NOT saved in the **Data repository** and is therefore available for information that is temporary or only valid on the specific device for which it was defined.



NOTE

The user can enter any UTF-8 character

9.2 General device settings

Overview of all standard commands

Standardized teach-in commands and reset commands can be sent via the **Standard Command**.

Table 10: General device settings - *Standard Commands*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
2	0x02	-	Standard command	UInt	-	8 bits	wo		Teach-in commands 65 = Single value Teach 71 = Start multi value teach 72 = Stop multi value teach 79 = Abort teach-in sequence 220 = Remove Teach Object Reset Commands: 128 = Device Reset 129 = Application Reset 130 = Restore factory settings 228 = Reset diagnostic parameters

Teach-in commands

For details on Teach-in Commands, see [table 32](#)

Reset Commands

For details on Reset Commands, see [table 11](#)

Reset

Table 11: General device settings - Restore Factory Settings

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
2	0x02	-	Standard command	UInt	-	8 bits	wo		128 = Device Reset 129 = Application Reset 130 = Restore Factory Settings 228 = Reset diagnostic parameters

Device reset

The sensor performs a restart.

Application Reset

Reset all application-relevant parameters.

Restore factory settings

The sensor is reset to factory settings.

Reset diagnostic parameters

Resetting the diagnostic key figures (minimum & maximum temperature (ISDU 153/4,153/5), operating hours counter (ISDU 190/2).

Locking functions

Table 12: General device settings - Device Access Locks

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
12	0x0C	-	Device access locks (Key lock)	Record	yes	2 bytes	rw	0	0 = Data Storage & LocalUserInterface available 2 = Data Storage locked 8 = Local User Interface locked 10 = Data Storage & LocalUserInterface locked

Various functions of a sensor can be locked or unlocked with **Device Access Locks**.

Bit 1 = **Data Storage**

Bit 3 = **Key Lock**

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
Bit 15 Bit 7 Bit 0

0 dec = **Data Storage & LocalUserInterface available**: Data storage and key lock enabled

2 dec = **Data Storage locked**: Data storage blocked

8 dec = **Local User Interface locked**: Key lock disabled

10 dec = **Data Storage & LocalUserInterface locked**: Data storage and control panel locked

Table 13: General device settings - Key Lock Type

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/rangeT
Index		Sub-Index							
DEC	HEX								
160	0xA0	-	Key Lock Type	UInt	yes	1 byte	rw	0	0 = Interface fully locked 1 = Teach-in available 2 = reserved

The **Key Lock Type** defines the available configuration options for the operating elements when the key lock is active:

0 = **Interface fully locked**: completely blocked

1 = **Teach-in available**: Teach-in available

Process data

Table 14: General device settings - *Process data select*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
Dec	Hex								
120	0x78	-	Process data select	UInt	yes	1 byte	rw	0	0 = Evaluation Mode 1 = Measurement Mode

The process data content that the sensor is to output cyclically can be determined via **Process data select**.

Evaluation Mode output of the CMV values (color match)

Measurement Mode: Measured value output of the current color

Table 15: Installation/Diagnostics - *Measurement Color Space*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
296	0x128	-	Measurement Color Space	UInt	yes	1 byte	rw		0 = RGB 1 = LAB

Setting the type of color measurement value - applies to the setting **Measurement Mode** in ISDU 120 (for details see [section 8](#)).

Table 16: Installation/Diagnostics - *Process Data Input / Output*

ISDU		Name	Data type	Data repository	Length	Access	Default value	Value/range	
Index									Sub-Index
DEC	HEX								
40	0x28	Process Data Input	PD in	-	12 bytes	ro	-	Referring to Process Data PD-in	

With the ISDU 40, the process data can be queried acyclically.

Device configuration

Table 17: General device settings - *Sender configuration*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
97	0x61	-	Sender configuration	Uint	-	1 byte	rw	0	0 = Sender active 1 = Sender not active

This ISDU can be used to switch off the Send LED.

If the transmitter is deactivated, **PD invalid** is output (process data invalid), all other values are 0.

Table 18: General device settings - Pin 2 configuration

ISDU			Name	Data type	Data Repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
121	0x79	-	Pin2 configuration	Uint	yes	1 byte	rw	39	0 = Deactivated 1 = External input (Smart Task) 20 = Blanking 34 = Switching output QL2 39 = Switching output QL1 80 = Single value teach QL1 81 = Single value teach QL2 90 = Activate job bank (LSB)

Pin 2 in the device connector plug (or white wire if connecting cable is used) can be assigned various output functions via **Pin 2 configuration**.

Deactivated	The signal level at pin 2 is not evaluated.
External input (Smart Task)	Input function for "Smart Task" control
Blanking	Blanking
	During blanking, all values in the process data remain frozen.
Switching output QL2 + Switching output QL1	Digital outputs for QL1 + QL2
Single value teach QL1 + Single value teach QL2	Single teach-in for QL1 + QL2
Activate job bank (LSB)	Activation of a job bank (LSB)

**NOTE**

With the 8-pin variant, the function of pin 1 is configured with ISDU 122.

Table 19: General device settings - Pin 5 configuration

ISDU			Name	Data type	Data storage	Length	Access	Default value	Value/Range
Index		Sub-index							
DEC	HEX								
122	0x7A	-	Pin5 configuration	Uint	no	8 bits	rw	1	0 = Deactivated 1 = External input (Smart Task) 20 = Blanking 80 = Single value teach QL1 81 = Single value teach QL2 90 = Activate job bank (MSB)

Configuration of the MF input on pin 5:

Deactivated	Signal level at pin 5 is not evaluated.
External input (Smart Task)	Input function for "Smart Task" control
Blanking	During blanking, all values in the process data remain frozen.
Single value teach QL1 + Single value teach QL2	Single teach-in for QL1 + QL2
Activate job bank (MSB)	Activation of a job bank (MSB)

Device search

Table 20: Device identification - *Find me*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
204	0xCC	-	Find me	UInt	no	8 bits	rw	0	0 = Find Me deactivated 1 = Find Me activated, yellow LED blinks with 1Hz

A connected sensor can be clearly identified via **Find me**. For machines with several identical sensors, it is therefore possible to uniquely identify the device with which communication is currently taking place.

When **Find me** is activated, the yellow LED of the sensor flashes at 1 Hz.

Event configuration

Table 21: General device settings - *Notification Handling*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
227	0xE3	-	Notification Handling	Uint	yes	1 byte	rw	0	0 = All enabled 1 = All disabled 2 = Events enabled, PD invalid flag disabled 3 = Events disabled, PD invalid flag enabled

In **Notification handling**, the generation of IO-Link events and the invalidity flag (status byte: bit 5 (table 5)) of the process date can be activated / deactivated in the sensor.

Display configuration

Table 22: General device settings - *Display settings*

ISDU			Name	Data type	Data repository	Length	Access	Default	Value/range
Index		Sub-Index							
DEC	HEX								
234	0xEA		Display settings	Record	yes	2 bytes	rw	-	
		1	Energy saving mode	UInt		8 bits		1	0 = OFF 1 = ON
		2	Turn display	UInt		8 bits		0	0 = Not turned 1 = Turned
Energy saving mode			When energy-saving mode is activated, the display is switched off 20 seconds after the last input. The display is reactivated by pressing any button.						
Turn display			Rotation of the display by 180°.						

Device information

Table 23: Device information - *Hardware Variant*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
440	0x1B8	-	Hardware Variant	UInt	-	1 byte	ro		0 = Standard IO 1 = Advanced IO 2 = RS-485

The hardware variant can be used to read out the interfaces of the connected variant:

0 = **Standard IO** (M12/5-pin with 2 x IO)

1 = **Advanced IO** (M12/8-pin with 5 x IO)

2 = **RS-485** (M12/8-pin with **RS485** + 3 x IO)

Table 24: Device identification - *Device ID Setup*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
16000	0x3E80	-	Device ID Setup	UInt	-	4 bytes	rw	8389262	8389262

The **IO-Link Device ID** can be selected here for initialization when establishing a connection.

9.3 Teach-in/Detection settings

Operating settings

Table 25: Teach-in/detection - *Operating Mode*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
110	0x6E	-	Operating Mode	UInt	yes	1 byte	rw	0	0 = Mark / Object positioning 1 = Fast sorting 2 = Object separation 3 = Color verification

Selection of the preset that the sensor should use for color evaluation. This results in the type of evaluation (see [table 29](#)) and the response time (see [table 27](#)).

Mark / Object positioning	Response time = no averaging CMV threshold = 900 Output Mode = Standard
Fast sorting	Response time = low averaging CMV threshold = 800 Output Mode = Best Fit Mode
Object separation	Response time = high averaging CMV threshold = 800 Output Mode = Best Fit Mode
Color verification	Response time = highest averaging CMV threshold = 950 Output Mode = Standard

Table 26: Teach-in/detection - *Currently selected operating mode*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
83	0x53	-	Currently selected operating mode	UInt	-	1 byte	ro	0	0 = Mark / Object positioning 1 = Fast sorting 2 = Object separation 3 = Color verification 255 = Manual setting

Read out the currently selected operating mode according to the setting in ISDU 110.

Table 27: Teach-in/detection - *Measurement averaging*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
89	0x59	0	Measurement averaging	UInt	yes	8 bits	rw	0	0 = No averaging 1 = Low averaging 2 = Medium averaging 3 = High averaging values 4 = Highest averaging

The filtering of the measured value can be set via **Measurement averaging**. The more values are included in the calculation, the lower the noise of the signal. However, this is at the expense of the sensor's response (latency).

Table 28: Response times CSS HighSpeed versus HighResolution

CSS HighSpeed	CSS HighResolution
0 = 36 µs	0 = 120 µs
1 = 72 µs	1 = 1 ms
2 = 150 µs	2 = 7.7 ms
3 = 300 µs	3 = 62 ms
4 = 600 µs	4 = 500 ms

Table 29: *Output Mode*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
293	0x125	-	Output Mode	UInt	yes	8 bits	rw	0	0 = Standard 1 = Best Fit Mode 2 = Coded Mode

Output Mode:

Standard: each output switches according to its individual setting (teach-in and sensitivity).

Best Fit Mode: only one output (Qint) is active if several colors are taught in. The quint with the closest match to the currently measured color is set as active.

Color Mode: such as **Best Fit Mode**. In addition, the number of the active Qint is output as a binary code on the QLs.

Table 30: *Sensing Distance Compensation*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
294	0x126	-	Distance regulation	UInt	yes	8 bits	rw	1	0 = Deactivated 1 = Activated

Distance regulation: Sensing range compensation (only for High Resolution models)

Table 31: *Color Mode*

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
295	0x127	-	Color Mode	UInt	yes	8 bits	rw	0	0 = C + I mode 1 = C mode

C + I mode: both the color and its intensity are considered here

C mode: evaluation of the color without its intensity (no gray levels distinguishable)

Teach-in

Table 32: Teach-in/detection - Teach-in Commands

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
2	0x02	-	Standard command	UInt	-	8 bits	wo		65 = Single Value Teach-in 71 = Start Multi Value Teach 72 = Stop Multi Value Teach 79 = Abort Teach-in sequence 220 = Remove Teach Object

The selection of the Qint for which the teach-in is to be carried out is determined via ISDU 58 (Teach-in channel).

Single Value Teach-in	Static teach-in of a color using Single Value Teach-in
Start Multi Value Teach-in + Stop Multi Value Teach-in	This function can be used to teach in the color to be detected while the process is running. To do this, the teach-in process is started by writing the value 71. From this point on, the sensor records the color values and interprets them. Write the value 72 to end the dynamic teach-in process.
Abort Teach-in sequence	Writing the value 79 interrupts a running teach-in process (Multi Value Teach-in).
Remove Teach Object	Deletes the teach-in values of the currently selected Qint / Teach-in channel (ISDU 58).

Table 33: Teach-in/detection - Teach-in channel / Teach state

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
58	0x3A	-	Teach-in channel	UInt	-	1 byte	rw	0	0 = Qint.1 1 = Qint.2 2 = Qint.3 ... 23 = Qint.24
59	0x3B	-	Teach-in status	UInt	-	1 byte	ro	0	0 = Idle 3 = SP12 success 4 = Wait for command 5 = Busy 7 = Error

Selection of the Qint switching output for the next teach-in via the system command.

The **Teach-in status** shows the current status of the teach-in process.

A teach-in can only be sent in the status **Idle** and **Error**.

The status always refers to the Qint. channel currently selected in **Teach-in channel** (ISDU 58).

Table 34: Teach-in/detection - Quality of Teach

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
114	0x72	-	Quality of Teach	UInt	-	1 byte	ro	-	0 ... 100 [in %]

The teach-in quality depends on the teach-in type:

- **Single value teach-in** always 100%
- **Multi value teach-in** decreasing value with increasing distribution of the taught-in color spectrum

Quality of Run

Table 35: Teach-in - Quality of Run

ISDU			Name	Data type	Data storage	Length	Access	Default value	Value/Range
Index		Sub-index							
DEC	HEX								
175	0xAF		Quality of Run	UInt	yes	1 byte	rw	0	0 ... 100

Quality of Run shows the process quality resulting from the current measured color value in relation to the set sensitivity and the maximum CMV value 999. Value in %.

ISDU 297 (see [table 37](#)) specifies this value specifically for each taught-in color. ISDU 175 outputs the minimum of all quality values as the worst-case scenario.

Table 36: Teach-in - Quality of run alarm, switching threshold

ISDU		Name	Data type	Data storage	Length	Access	Default value	Value/Range	
Index									Sub-index
DEC	HEX								
176	0xB0	Quality of run alarm threshold	Uint	yes	1 byte	rw	50	0 ... 90	

Alarm threshold for the process quality (ISDU 175) - if the value falls below the set value, a status bit (see [table 5](#)) is set and an IO-Link event (see [table 60](#)) is output. Value in %.

Table 37: Quality Levels

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
297	0x129	-	Quality Levels	Array 24 x UINT8	-	24 bytes	ro		

Quality Levels shows the process quality of all 24 color banks.

Teach-in data 1 to 24

Here are the teach-in data for the associated Qint. Outputs are stored.

The teach-in data contains all relevant information on the colors stored in each case. This means that each taught-in color can be transferred to another sensor, for example.

Table 38: Teach-in data - Qint.

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
			Qint. teach data	Record		44 bytes			

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
60	0x3C	1	Nucleus L	Float	yes	4 bytes	rw		
62	0x3E	2	Nucleus a	Float	yes	4 bytes	rw		
16384	0x4000	3	Nucleus b	Float	yes	4 bytes	rw		
16386	0x4002	4	Rotation L	Float	yes	4 bytes	rw		
16388	0x4004	5	Rotation a	Float	yes	4 bytes	rw		
16390	0x4006	6	Rotation b	Float	yes	4 bytes	rw		
16392	0x4008	7	Extent L	Float	yes	4 bytes	rw		
16394	0x400A	8	Extent A	Float	yes	4 bytes	rw		
16396	0x400C	9	Extent B	Float	yes	4 bytes	rw		
16398	0x400E	10	Application Hysteresis R	UInt	yes	2 bytes	rw		
16400	0x4010	11	Application Hysteresis G	UInt	yes	2 bytes	rw		
16402	0x4012	12	Application Hysteresis B	UInt	yes	2 bytes	rw		
16404	0x4014	13	Signal Damping	UInt	yes	2 bytes	rw		
16406	0x4016								
16408	0x4018								
16410	0x401A								
16412	0x401C								
16414	0x401E								
16416	0x4020								
16418	0x4022								
16420	0x4024								
16422	0x4026								
16424	0x4028								
16426	0x402A								

Qint. teach data for Qint. 1 to Qint. 24:

ISDU 60 = Qint. 1 teach data

ISDU 62 = Qint. 2 teach data

ISDU 16384 = Qint. 3 teach data

...

ISDU 16426 = Qint. 24 teach data

Nucleus: Describes the center point of the teach-in object as a LAB triple

Rotation: Describes the rotation of the extension of the teach-in object around the core as a LAB vector

Extent: Describes the extent of the rotated teach-in object as a 3D vector

Application Hysteresis: Describes the signal noise measured during teach-in

Signal Damping: Selected signal attenuation for teach-in objects

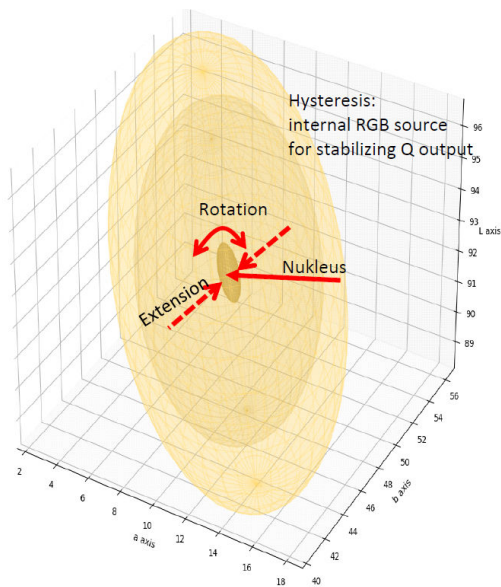


Figure 4: Signal noise

Table 39: Teach-in data - Qint.

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
			Qint. configuration	Record		3 bytes			
61 63	0x3C 0x3E	1	Referenced Teach Object	UInt	yes	1 byte	rw	0	0 ... 31
16385	0x4000	2	Sensitivity	UInt	yes	2 bytes	rw	900	0 ... 999
16387	0x4002								
16389	0x4004								
16391	0x4006								
16393	0x4008								
16395	0x400A								
16397	0x400C								
16399	0x400E								
16401	0x4010								
16403	0x4012								
16405	0x4014								
16407	0x4016								
16409	0x4018								
16411	0x401A								
16413	0x401C								
16415	0x401E								
16417	0x4020								
16419	0x4022								
16421	0x4024								
16423	0x4026								
16425	0x4028								
16427	0x402A								

Qint. configuration for Qint. 1 to Qint. 24:

ISDU 61 = Qint. 1 configuration

ISDU 63 = Qint. 2 configuration

ISDU 16385 = Qint. 3 configuration

...

ISDU 16427 = Qint. 24 configuration

Referenced Teach Object: Assignment of the sensitivity threshold for the associated teach-in object.

Sensitivity: Sensitivity value in relation to the degree of match (CMV), determines the color match at which the respective Qint switches

- 0 = no match (Qint always ON)
- 999 = full match

Table 40: Teach-in / Names of Qint.

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
4081	0xFF1	-	Names Quint. 1 to 16	Record	-	192 bytes		-	Color 1 ... 16
		1	Name Quint. 1	String	yes	12 bytes	rw		
		2	Name Quint. 2	String	yes	12 bytes	rw		
		3	Name Quint. 3	String	yes	12 bytes	rw		
		4	Name Quint. 4	String	yes	12 bytes	rw		
		5	Name Quint. 5	String	yes	12 bytes	rw		
		6	Name Quint. 6	String	yes	12 bytes	rw		
		7	Name Quint. 7	String	yes	12 bytes	rw		
		8	Name Quint. 8	String	yes	12 bytes	rw		
		9	Name Quint. 9	String	yes	12 bytes	rw		
		10	Name Quint. 10	String	yes	12 bytes	rw		
		11	Name Quint. 11	String	yes	12 bytes	rw		
		12	Name Quint. 12	String	yes	12 bytes	rw		
		13	Name Quint. 13	String	yes	12 bytes	rw		
		14	Name Quint. 14	String	yes	12 bytes	rw		
		15	Name Quint. 15	String	yes	12 bytes	rw		
		16	Name Quint. 16	String	yes	12 bytes	rw		

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
4082	0xFF2	-	Names Quint. 17 to 24	Record	-	96 bytes		-	Color 17 ... 24
		1	Name Quint. 17	String	yes	12 bytes	rw		
		2	Name Quint. 18	String	yes	12 bytes	rw		
		3	Name Quint. 19	String	yes	12 bytes	rw		
		4	Name Teach Object 20	String	yes	12 bytes	rw		
		5	Name Quint. 21	String	yes	12 bytes	rw		
		6	Name Quint. 22	String	yes	12 bytes	rw		
		7	Name Quint. 23	String	yes	12 bytes	rw		
		8	Name Quint. 24	String	yes	12 bytes	rw		

A name can be assigned to each teach-in object.

9.4 Installation/Diagnostics

Reading all color match values (CMV)

Table 41: Installation/Diagnostics - Color Match Values

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
165	0xA5	-	Color Match Values	Record		48 bytes		-	0 ... 999
		Byte 47-48	CMV 1	UInt	-	2 bytes	ro		0 ... 999
		Byte 45-46	CMV 2	UInt	-	2 bytes	ro		0 ... 999
		Byte 43-44	CMV 3	UInt	-	2 bytes	ro		0 ... 999
		Byte 41-42	CMV 4	UInt	-	2 bytes	ro		0 ... 999
		Byte 39-40	CMV 5	UInt	-	2 bytes	ro		0 ... 999
		Byte 37-38	CMV 6	UInt	-	2 bytes	ro		0 ... 999
		Byte 35-36	CMV 7	UInt	-	2 bytes	ro		0 ... 999
		Byte 33-34	CMV 8	UInt	-	2 bytes	ro		0 ... 999
		Byte 31-32	CMV 9	UInt	-	2 bytes	ro		0 ... 999
		Byte 29-30	CMV 10	UInt	-	2 bytes	ro		0 ... 999
		Byte 27-28	CMV 11	UInt	-	2 bytes	ro		0 ... 999
		Byte 25-26	CMV 12	UInt	-	2 bytes	ro		0 ... 999
		Byte 23-24	CMV 13	UInt	-	2 bytes	ro		0 ... 999
		Byte 21-22	CMV 14	UInt	-	2 bytes	ro		0 ... 999
		Byte 19-20	CMV 15	UInt	-	2 bytes	ro		0 ... 999
		Byte 17-18	CMV 16	UInt	-	2 bytes	ro		0 ... 999
		Byte 15-16	CMV 17	UInt	-	2 bytes	ro		0 ... 999
		Byte 13-14	CMV 18	UInt	-	2 bytes	ro		0 ... 999
		Byte 11-12	CMV 19	UInt	-	2 bytes	ro		0 ... 999
		Byte 9-10	CMV 20	UInt	-	2 bytes	ro		0 ... 999
		Byte 7-8	CMV 21	UInt	-	2 bytes	ro		0 ... 999
		Byte 5-6	CMV 22	UInt	-	2 bytes	ro		0 ... 999
		Byte 3-4	CMV 23	UInt	-	2 bytes	ro		0 ... 999
		Byte 1-2	CMV 24	UInt	-	2 bytes	ro		0 ... 999

The CMV shows the degree of compliance with the individual teach-in objects:

0 = no match

999 = full match

System status

Table 42: Device State

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
36	0x24	-	Device Status	Uint	-	1 byte	ro	0	0 = Device is OK 1 = Maintenance required 2 = Out of specification ¹⁾ 3 = Functional check ¹⁾ 4 = Failure ¹⁾

¹⁾ Not supported by the device

Device is OK: Device is in order

Maintenance required : Action required, e.g. teach-in

Out of specification: Out of specification, not supported by the device

Functional check: Function test, not supported by the device

Failure: Error status, not supported by the device

Table 43: Installation/Diagnostics - Device status

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
37	0x25	-	Detailed device status	Array	-	15 bytes	ro	-	-
			Event 1			3 bytes			OxE48CA4 = Warning appeared - Quality of run alarm OxE48CA0 = Warning appeared - Short circuit on output pin OxE48CA8 = Warning appeared - Alarm upper temperature threshold OxE48CAB = Warning appeared - Alarm operating hours OxE48CAF = Warning appeared - Alarm lower temperature threshold OxF45000 = Error appeared - Device hardware fault 0x000000 = no event appeared
			Event 2			3 bytes			
			Event 3			3 bytes			
			Event 4			3 bytes			
			Event 5	-		3 bytes			

The detailed **Device Status** shows up to 5 events of the warning and error type that are currently present. The information is available as a data string of 5 x 3 bytes, in the order in which it occurs.

Byte 1: Event qualifier, [section 11.1](#)

Byte 2 + 3: Event code, [section 11.2](#)

Condition monitoring

Table 44: Installation/Diagnostics - Temperature

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
153	0x99	0	Temperature	Record	-	5 bytes	ro	-	
		1	Current temperature	INT		8 bits			-127 ... 127 °C
		2	Max. temperature all time	INT		8 bits			-127 ... 127 °C
		3	Min. temperature all time	INT		8 bits			-127 ... 127 °C
		4	Max. temperature since last reset	INT		8 bits			-127 ... 127 °C
		5	Min. temperature since last reset	INT		8 bits			-127 ... 127 °C

Read out the operating temperature of the sensor. The values of **Max. temperature since last reset** and **Min. temperature since last reset** are deleted via the **Standard command Reset diagnostic parameters** (index 2, value 228).

Table 45: Installation/Diagnostics - Alarm thresholds for diagnostic parameters

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
179	0xB3		Alarm thresholds for diagnostic parameters	Record	yes	6 bytes	rw	-	
		1	Upper Temperature Threshold	INT		1 byte		80	
		2	Lower Temperature Threshold	INT	yes	1 byte		-30	
		3	Operating Hours Threshold	INT	yes	4 bytes		40000	

The **Parameter Alarm threshold for diagnostic parameters** offers the option of defining alarm thresholds for certain diagnostic values provided by the device. If these alarm thresholds are exceeded or not reached, a corresponding event is generated.

Table 46: Installation/Diagnostics - Operating hours

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
190	0xBE	0	Operating hours	Record	-	8 bytes	ro	-	
		1	Total operating hours	UInt		32 bits (Offset 32 bits)			0 ... 1000000
		2	Operating hours since last reset	UInt		32 bits (Offset 0 bits)			0 ... 1000000

The **Total operating hours** parameter displays how many total hours (h) the device has already been in operation. This value cannot be reset.

Parameter Operating hours since last reset displays how many hours (h) the device has been in operation since the last reset of the diagnostic parameters. The diagnostic parameters are reset using the **Reset diagnostic parameter** standard command (index 2, value 228).

Table 47: Teach-in/detection - **Standard command**

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-index							
DEC	HEX								
2	0x02	-	Standard command	UInt	-	1 byte	wo	-	228 = Reset diagnostic parameter

Resetting the minimum and maximum temperature (ISDU 1534/4, 153/5) and the operating hours (ISDU 190/2) since the last reset.

Table 48: Installation/Diagnostics - **Distance to Object**

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
229	0xE5		Distance to Object	Record	-	3 bytes	ro	-	
		1	Distance	UInt		2 bytes			0 ... 5000
		2	Distance Qualifier	UInt		1 byte			0 = Distance in range / valid 3 = No distance information / distance invalid

Distance information only available for the CSS High Resolution variants.

Distance: Unit = 1/10 mm, measured value only if available, otherwise the value is 0.

Distance Qualifier: 0 = distance value valid (within the working range matching the sensor), 3 = no distance information available

Distance information for the **CSS High Resolution** variants.

9.5 Smart Tasks

Smart Tasks process the various Smart Sensor signals for detection and measurement, linking them to binary switching signals from an external sensor if necessary. These signals can be imported via pin 2 (see Pin 2 configuration, ISDU 121). The Smart Task uses this data to generate the requisite process information – tailored to the task at hand in the plant. This saves time during data evaluation in the control, accelerates machine processes, and makes high-performance, cost-intensive additional hardware unnecessary.

- Decentralized signal analysis directly at the sensor
- Faster signal capture and processing
- Through Smart Tasks, Smart Sensors deliver the information that the plant process actually requires – no separate data preparation necessary in the control

9.5.1 Smart Task A10

Table 49: Smart Task - Inverter Ext. input

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1093	0x445	-	Inverter Ext.input	Record	yes	1 byte	rw		true = Inverted false = Not inverted
		1	Inverter Ext.input 1 (Pin 2 / Pin 1)	UInt	yes	1 bit	rw	0	true = Inverted false = Not inverted
		2	Inverter Ext.input 2 (Pin 5)	UInt	yes	1 bit	rw	0	true = Inverted false = Not inverted

The **Inverter Ext. input** function allows you to change the signal logic of a pin 2 / pin 5 configured as **Ext. input**.

The setting is made according to the bit pattern:

'00000000'

Bit 0 = Inverter external input 2 (pin 5), default = 0

Bit 1 = Inverter external input 1 (pin 2), default = 0



NOTE

Pin 2 applies to the 5-pin version. In the 8-pin version, this is pin 1.

Table 50: Smart Task - SLTI Version

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1208	0x4B8	-	SLTI Version	String	-	8 bytes	ro	1.1.0	

The version identifier can be read out with the ISDU 1208.

The SLTI version contains the version number of the Smart Task basic logic.

1.1.0: 3-digit version identifier, with dot separation (UTF-8 string)

Table 51: Smart Task - Input Selector

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1209 1214 1219 1224	0x4B9 0x4BE 0x4C3 0x4C8	-	Input Selector 1 ... 4	Record	yes	4 bytes	rw		
		1	Qint. 1	UInt	yes	1 bit	rw	1	true = Qint.1 selected false = Qint.1 not selected
		2	Qint. 2	UInt	yes	1 bit	rw	0	true = Qint.2 selected false = Qint.2 not selected
		3	Qint. 3	UInt	yes	1 bit	rw	0	true = Qint.3 selected false = Qint.3 not selected
		4	Qint. 4	UInt	yes	1 bit	rw	0	true = Qint.4 selected false = Qint.4 not selected
		5	Qint. 5	UInt	yes	1 bit	rw	0	true = Qint.5 selected false = Qint.5 not selected
		6	Qint. 6	UInt	yes	1 bit	rw	0	true = Qint.6 selected false = Qint.6 not selected
		7	Qint. 7	UInt	yes	1 bit	rw	0	true = Qint.7 selected

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
									false = Qint.7 not selected
		8	Qint. 8	UInt	yes	1 bit	rw	0	true = Qint.8 selected false = Qint.8 not selected
		9	Qint. 9	UInt	yes	1 bit	rw	0	true = Qint.9 selected false = Qint.9 not selected
		10	Qint. 10	UInt	yes	1 bit	rw	0	true = Qint.10 selected false = Qint.10 not selected
		11	Qint. 11	UInt	yes	1 bit	rw	0	true = Qint.11 selected false = Qint.11 not selected
		12	Qint. 12	UInt	yes	1 bit	rw	0	true = Qint.12 selected false = Qint.12 not selected
		13	Qint. 13	UInt	yes	1 bit	rw	0	true = Qint.13 selected false = Qint.13 not selected
		14	Qint. 14	UInt	yes	1 bit	rw	0	true = Qint.14 selected false = Qint.14 not selected
		15	Qint. 15	UInt	yes	1 bit	rw	0	true = Qint.15 selected false = Qint.15 not selected
		16	Qint. 16	UInt	yes	1 bit	rw	0	true = Qint.16 selected false = Qint.16 not selected
		17	Qint. 17	UInt	yes	1 bit	rw	0	true = Qint.17 selected false = Qint.17 not selected
		18	Qint. 18	UInt	yes	1 bit	rw	0	true = Qint.18 selected false = Qint.18 not selected
		19	Qint. 19	UInt	yes	1 bit	rw	0	true = Qint.19 selected false = Qint.19 not selected
		20	Qint. 20	UInt	yes	1 bit	rw	0	true = Qint.20 selected false = Qint.20 not selected
		21	Qint. 21	UInt	yes	1 bit	rw	0	true = Qint.21 selected false = Qint.21 not selected
		22	Qint. 22	UInt	yes	1 bit	rw	0	true = Qint.22 selected false = Qint.22 not selected
		23	Qint. 23	UInt	yes	1 bit	rw	0	true = Qint.23 selected false = Qint.23 not selected
		24	Qint. 24	UInt	yes	1 bit	rw	0	true = Qint.24 selected false = Qint.24 not selected
		25	Ext.input 1	UInt	yes	1 bit	rw	0	true = Ext.input 1 selected false = Ext.input 1 not selected
		26	Ext.input 2	UInt	yes	1 bit	rw	0	true = Ext.input 2 selected false = Ext.input 2 not selected

For programming QL1 ... QL4: Selection of the Qint. to be linked according to bit pattern:

0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1
 Bit 31 Bit 23 Bit 15 Bit 7 Bit 0

Table 52: Smart Task - Logic

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1210	0x4BA	-	Logic 1 ... 4	UInt		1 byte	rw	2	1 = And 2 = Or
1215	0x4BF								
1220	0x4C4								
1225	0x4C9								

For logical linking of the selected Qint in Input Selector 1.

Table 53: Smart Task - Timer

ISDU		Sub-Index	Name	Data type	Data repository	Length	Access	Default value	Value/range
Index									
DEC	HEX								
1211	0x4BB	-	Timer 1 ... 4	UInt		1 byte	rw	0	0 = Deactivated 1 = T-on delay 2 = T-off delay 3 = T-on/T-off delay 4 = Impulse
1216	0x4C0								
1221	0x4C5								
1226	0x4CA								

For setting a time function for QL1...4.

T-on delay
N delay

T-off delay
OFF switching delay

T-on/T-off delay
N/OFF switching delay

Impulse
fixed pulse time

Table 54: Smart Task - Time Setup

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1212	0x4BC	-	Time Setup 1 ... 4	UInt		2 bytes	rw	1	0 ... 30000 = Time value in ms
1217	0x4C1								
1222	0x4C6								
1227	0x4CB								

For setting the desired time for the time function.

1 ... 30,000 in ms, factory setting = 1 ms

Table 55: Smart Task - Inverter

ISDU			Name	Data type	Data repository	Length	Access	Default value	Value/range
Index		Sub-Index							
DEC	HEX								
1213	0x4BD	-	Inverter 1 ... 4	UInt		1 byte	rw	0	0 = Not inverted 1 = Inverted
1218	0x4C2								
1223	0x4C7								
1228	0x4CC								

To set the inversion of QL1...4.

0 = Not inverted

1 = Inverted

10 Sensor replacement/data storage

All **IO-Link Device** have a backup and restore functionality - **Data Storage (DS)**. Thanks to the **IO-Link Data Storage** function, previous parameters can be saved and transferred to the replacement device, eliminating the need to re-parameterize the replacement device.

In the world of automation, there are several names for the same function:

- Data storage
- Backup and restore
- Parameter server
- Device replacement



NOTE

The **DS** function was introduced with the **IO-Link** specification V1.1.

It is therefore possible that devices older than 2013 were programmed according to the **IO-Link** specification V1.0 and do not support the entire feature.

Data storage function

Replacing a **IO-Link Device** requires the replacement hardware and the application-specific parameterization of the device.

When data storage is activated, the **IO-Link Master** always saves the last valid setting parameters of all connected **IO-Link Device** in its local memory. If one of the connected **IO-Link Devices** is replaced with a functionally compatible replacement device, the **IO-Link Master** automatically transfers the last valid parameter set of the predecessor sensor to the new **IO-Link Device**.

When activating the new **IO-Link Device** in the **IO-Link Master**, you can choose between different **IO-Link Master** behaviors:

- **NONE**: No backup of the device parameters is made in the **IO-Link Master**.
- **BACKUP/RESTORE**: The **IO-Link Master** saves the parameterization of the connected **IO-Link Devices** (initially automatically) and is ready to restore it in the event of a device replacement.
- **RESTORE**: No automatic backup of the device parameters is performed in the **IO-Link Master**. Manually initiated backups are possible. The **IO-Link Master** monitors the port and restores the parameterization when a replacement device is detected.



NOTE

- To be able to use data storage, it must be activated in the **IO-Link Master**.
 - If the changeover of one or more sensor parameters is initiated via the control unit, the control unit must activate the so-called **Data Storage Upload Request-Flag** as the final command in the sensor. Only this initiates the data repository.
 - Depending on the data volume and the **IO-Link Master** used, the upload/download of sensor parameters via the data storage function can take between a few hundred milliseconds and up to three seconds (typical values; values may vary in individual cases).
 - For details on the use of data storage, see "IO-Link Interface and System Specification, V1.1.2, Chapter 10.4 Data Storage (DS)" under www.io-link.com, menu item Downloads.
 - Parameters that do not participate in data storage are marked in the **IODD** overview.
-



NOTE

To set the **DSfunction**, read the manufacturer-specific instructions for the **IO-Link Master**.

Port 1 - Pin 4

Figure 5: Web server user interface of a SIG350 from SICK

Validation

The **Validation** function is always available in conjunction with the **DS** functionality, even if **Data Storage** is not being used.

When replacing a device, it must be ensured that it is replaced by a compatible device. For this reason, the **DS** functionality includes a check for the correct device.

For this purpose, the **Device ID** and **Vendor ID** of the connected device are compared with the ID data in the **DS**.

If the IDs match the IDs saved in the backup of the **IO-Link Masters**, the **IO-Link Master** continues and compares the parameterization of the device with those saved in the **DS**.

If the IDs do not match, an error event is triggered to inform the above-mentioned communication layers of the mismatch:

Table 56: **EventCodes** for ports¹⁾

EventCode ID	Definition and recommended maintenance action	Type
0x0000 to 0x17FF	Reserved	
0x1800	No device (communication) Trigger: SMI_PortEvent (0x1800) by SM_PortMode (COMLOST)	Error
0x1801	Startup parametrization error - check parameter	Error
0x1802	Incorrect Vendor ID - Inspection Level mismatch Trigger: SM_PortMode (COM_FAULT)	Error
0x1803	Incorrect Device ID - Inspection Level mismatch Trigger: SM_PortMode (COM_FAULT)	Error

Backup

Depending on the data backup behavior setting (**BACKUP/RESTORE**), **IO-Link Master** automatically creates the first backup (**Backup**) of the application-specific parameterization.

If all settings on the device are correct, the backup is complete.

In most cases, however, adjustments are necessary: the content of **Backup** must then be updated. With the selection **RESTORE**, the **Backup** must always be created manually.

There are various options for parameterizing **IO-Link Devices**. There are therefore various options for updating the content of the **Backup**.

¹⁾ Source: IO-Link Interface and System Specification V1.1.3, June 2019

DS update options

- Parameter changes on the device itself (display, buttons, potentiometer, etc.):
All changes made directly to the device are automatically reported to **IO-Link Master**. The **IO-Link Device** informs the **IO-Link Master** about parameter changes and the **IO-Link Master** updates its data memory accordingly.

**NOTICE**

The **Backup** is not updated in **RESTORE** mode!

- Parameter changes via PLC (also SICK function blocks) or other programming tools:
If PLC programs or other application-specific processes access the device via the IO-Link interface and change parameters, no automatic action takes place. The user program itself must ensure that the content of the data memory is updated after the adjustments. The function modules offered by SICK to simplify the exchange of service data do not trigger **Backup** updates. System commands for controlling the data storage functionality are available for this purpose.

Table 57: System commands for data storage functionality

Command (hex)	Command (dec)	Command name	M/O	Definition
0x00	0	Reserved		
0x01	1	ParamUploadStart	0	Start parameter upload
0x02	2	ParamUploadEnd	0	Stop parameter upload
0x03	3	ParamDownloadStart	0	Start parameter download
0x04	4	ParamDownloadEnd	0	Stop parameter download
0x05	5	ParamDownloadStore	0	Finalize parameterization and start data storage

If the changes made by the PLC communication are to be recorded in the DS, the **IO-Link Device** must signal this to the **IO-Link Master** via an IO-Link event.

This event is generated by the device when it receives the system command **ParamDownloadStore** (index 0x02, value 0x05).

→ To add new parameters to the **Backup** of the DS, write an ISDU with the value 0x05 to the **System Command** ISDU (index 2) at the end of your sequence.

- Engineering tools such as SOPAS (off-site commissioning):
Engineering tools are obliged to set the DS upload flag after parameterization has been completed via their graphical user interfaces. This applies both to engineering tools that use the IO-Link interface itself (note: new since V1.1.3) and if additional interfaces such as TCP/IP, USB are used.

**NOTE**

Since engineering tools did not behave in this way before **IO-Link Interface Version V1.1.3**, pay particular attention to whether the **IO-Link Master** updates its DS content when the **IO-Link Device** is connected back to the **IO-Link Master**.

- Cloud interfaces/Dual Talk (IIoT connections, such as REST API, MQTT or OPC UA):
Cloud interfaces from **IO-Link Master** often offer the option of changing parameters from **IO-Link Devices**. The same applies to these interfaces as for access by PLC programs - there is no automatic update. The respective DS update commands of the cloud protocol must be sent to **IO-Link Master**.

11 Events

IO-Link events

With **Events** (events), an **IO-Link Device** reports events to the **IO-Link Master** without being prompted to do so by the **IO-Link Master**.

Events are the only way for an **IO-Link Device** to report a sporadic event, information or problem. An **IO-Link Master** can use events to inform about port-specific events, e.g. the disconnection of an **IO-Link Device** from the **IO-Link Master**.



NOTE

Not all **IO-Link Master** support the event mechanism, especially older ones. You can deactivate the generation of events on the device page in “**Notification handling** (ISDU 227)”.

An event consists of:

- **Event Qualifier:** Specification of information about:
 - **Instance** (event instance)
 - **Source** (event source)
 - **Type** (event type)
 - **Mode** (event mode)
- **Event Code:** Details of the event content

11.1 Event Qualifier

The **Event Qualifier** (event qualifier) is a byte that contains some important information about the event.

Table 58: Event qualifier

Mode		Type		Source	Instance		
Bit 7	6	5	4	3	2	1	Bit 0
0 = reserved		0 = reserved		0 = Device	0 = unknown		
1 = Event single shot		1 = Notification		1 = Master	1 ... 3 = reserved		
2 = Event disappears		2 = Warning			4 = Application		
3 = Event appears		3 = Error					

Type

Type (event type) are classified as follows:

Table 59: Type

Value	Definition	Description
0	Reserved	-
1	Notification	For information purposes only; the system is not restricted.
2	Warning	System is still functional, but impaired in some way. You must rectify the problem as quickly as possible by taking appropriate measures.
3	Error	The system is no longer functional. Depending on the cause of the error, it may be possible to restore the function.

11.2 Event Code

An event outputs a 2-byte long **Event Code** that contains the cause for the occurrence of the event.

The information on the event source from **Event Qualifier** can be used to differentiate where the event comes from.

11.2.1 Device-specific events

**NOTE**

Not all IO-Link masters support the event mechanism.

In **Notification Handling** (Index 227), the generation of events can be deactivated on the device side.

IO-Link Devices support manufacturer-specific **Event Codes**, which must be described in the documentation belonging to the **IO-Link Device**.

The following device-specific events are supported:

Table 60: Device-specific Events

Code		Name	Type	Note	Action
Dec	Hex				
20480	0x5000	Device hardware fault	Error	Fatal device error	Replacement required.
36000	0x8CA0	Short circuit on Qx	Warning	Triggered in the event of a short circuit on at least one digital output. Overcurrent detection.	Check device connection.
36001	0x8CA1	New parameters	Notification	Parameters have been amended (only when changing the sensing range using control elements on the sensor housing or using the external teach-in via pin 2).	None
36004	0x8CA4	Quality of run alarm	Warning	Operational safety alarm	Clean the optical surfaces (sensor and reflector).
36008	0x8CA8	Alarm upper temperature threshold	Warning	Upper temperature threshold has been exceeded.	Cool down sensor.
36011	0x8CAB	Alarm operating hours	Warning	Alarm threshold for operating hours reached	Prepare on-site service or device exchange.
36015	0x8CAF	Alarm lower temperature threshold	Warning	Lower temperature threshold has been exceeded.	Warm up sensor.

Example of device-specific Event Code

Sensor from SICK sends the event "Successful teach-in":

▶ LastEvent_Type	'Notification'	{...}		STRING
▶ LastEvent_Source	'Device'	{...}		STRING
▶ LastEvent_Qualifier	16#54		Hex	SINT
▶ LastEvent_Mode	'Event single shot'	{...}		STRING
▶ LastEvent_Instance	'Application'	{...}		STRING
▶ LastEvent_Code	16#1801		Hex	INT

Figure 6: Source: Logix Designer, Studio 5000 using SIG350 as IO-Link Master

Common **Event Codes** are defined in the IO-Link interface specification (Table D.1):

Table 61: Event Codes for IO-Link Devices²⁾

Event Code ID	Definition and recommended maintenance action	DeviceStatus Value	Type
0x0000	No malfunction	0	Notification
0x1000	General malfunction – unknown error	4	Error
0x1001 to 0x17FF	Reserved		

²⁾ Source: IO-Link Interface Specification V1.1.3, June 2019; Table D.1 - EventCodes for Devices

Event Code ID	Definition and recommended maintenance action	DeviceStatus Value	Type
0x1800 to 0x18FF	Vendor specific		
0x1900 to 0x3FF	Reserved		
0x4000	Temperature fault – Overload	4	Error
0x4001 to 0x420F	Reserved		
0x4210	Device temperature overrun – Clear source of heat	2	Warning
0x4211 to 0x421F	Reserved		
0x4220	Device temperature underrun – Insulate Device	2	Warning
0x4221 to 0x4FFF	Reserved		
0x5000	Device hardware fault – Device exchange	4	Error
0x5001 to 0x500F	Reserved		
0x5010	Component malfunction – Repair or exchange	4	Error
0x5011	Non volatile memory loss – Check batteries	4	Error
0x5012	Batteries low – Exchange batteries	2	Warning
0x5013 to 0x50FF	Reserved		
0x5100	General power supply fault – Check availability	4	Error
0x5101	Fuse blown/open – Exchange fuse	4	Error
0x5102 to 0x510F	Reserved		
0x5110	Primary supply voltage overrun – Check tolerance	2	Warning
0x5111	Primary supply voltage underrun – Check tolerance	2	Warning
0x5112	Secondary supply voltage fault (Port Class B) – Check tolerance	2	Warning
0x5113 to 0x5FFF	Reserved		
0x6000	Device software fault – Check firmware revision	4	Error
0x6001 to 0x631F	Reserved		
0x6320	Parameter error – Check data sheet and values	4	Error
0x6321	Parameter missing – Check data sheet	4	Error
0x6322 to 0x634F	Reserved		
0x6350	Reserved		
0x6351 to 0x76FF	Reserved		
0x7700	Wire break of a subordinate device – Check installation	4	Error
0x7701 to 0x770F	Wire break of subordinate device 1 ...device 15 – Check installation	4	Error
0x7710	Short circuit – Check installation	4	Error
0x7711	Ground fault – Check installation	4	Error

Event Code ID	Definition and recommended maintenance action	DeviceStatus Value	Type
0x7712 to 0x8BFF	Reserved		
0x8C00	Technology specific application fault – Reset Device	4	Error
0x8C01	Simulation active – Check operational mode	3	Warning
0x8C02 to 0x8C0F	Reserved		
0x8C10	Process variable range overrun – Process Data uncertain	2	Warning
0x8C11 to 0x8C1F	Reserved		
0x8C20	Measurement range exceeded – Check application	4	Error
0x8C21 to 0x8C2F	Reserved		
0x8C30	Process variable range underrun – Process Data uncertain	2	Warning
0x8C31 to 0x8C3F	Reserved		
0x8C40	Maintenance required – Cleaning	1	Warning
0x8C41	Maintenance required – Refill	1	Warning
0x8C42	Maintenance required – Exchange wear and tear parts	1	Warning
0x8C43 to 0x8C9F	Reserved		
0x8CA0 to 0x8DFF	Vendor specific		
0x8E00 to 0xAFFF	Reserved		
0xB000 to 0xB0FF	Reserved for Safety extensions		
0xB100 to 0xBFFF	Reserved for profiles		
0xC000 to 0xFF90	Reserved		
0xFF91	Data Storage upload request ("DS_UPLOAD_REQ") – internal, not visible to user	0	Notification (single shot)
0xFF92 to 0xFFAF	Reserved		
0xFFB0 to 0xFFB7	Reserved for Wireless extensions		
0xFFB8 to 0xFFFF	Reserved		

Example of common Event Code

Sensor from SICK transmits the event of a "short circuit" fault occurring:

▶ LastEvent_Type	'Error'	{...}	STRING
▶ LastEvent_Source	'Device'	{...}	STRING
▶ LastEvent_Qualifier	16#f4	Hex	SINT
▶ LastEvent_Mode	'Event appears'	{...}	STRING
▶ LastEvent_Instance	'Application'	{...}	STRING
▶ LastEvent_Code	16#7710	Hex	INT

Figure 7: Source: Logix Designer, Studio 5000 using SIG350 as IO-Link Master

11.2.2 Port-specific events

Port-specific events are events that are output by the **IO-Link-Master**. The reason for their occurrence has something to do with the port to which a device is connected.

A distinction is also made here between common **Event Codes** that are specified in the IO-Link interface specification (Table D.2) and events that are **IO-Link Master-specific**:

Table 62: *EventCodes* for ports ³⁾

EventCode ID	Definition and recommended maintenance action	Type
0x0000 to 0x17FF	Reserved	
0x1800	No Device (communication) Trigger: SMI_PortEvent (0x1800) by SM_PortMode (COMLOST)	Error
0x1801	Startup parametrization error – check parameter	Error
0x1802	Incorrect VendorID – Inspection Level mismatch Trigger: SM_PortMode (COMP_FAULT)	Error
0x1803	Incorrect DeviceID – Inspection Level mismatch Trigger: SM_PortMode (COMP_FAULT)	Error
0x1804	Short circuit at C/Q – check wire connection	Error
0x1805	PHY overtemperature – check Master temperature and load	Error
0x1806	Short circuit at L+ – check wire connection	Error
0x1807	Overcurrent at L+ – check power supply (e.g. L1+)	Error
0x1808	Device Event overflow	Error
0x1809	Backup inconsistency – memory out of range (2048 octets) Trigger: SMI_PortEvent (0x1809) by DS_Fault (SizeCheck_Fault)	Error
0x180A	Backup inconsistency – identity fault Trigger: SMI_PortEvent (0x180A) by DS_Fault (Identification_Fault)	Error
0x180B	Backup inconsistency – Data Storage unspecific error Trigger: SMI_PortEvent (0x180B) by DS_Fault (All other incidents)	Error
0x180C	Backup inconsistency – upload fault	Error
0x180D	Parameter inconsistency – download fault	Error
0x180E	P24 (Class B) missing or undervoltage	Error
0x180F	Short circuit at P24 (Class B) – check wire connection (e.g. L2+)	Error
0x1810	Short circuit at I/Q – check wiring	Error
0x1811	Short circuit at C/Q (if digital output) – check wiring	Error
0x1812	Overcurrent at I/Q – check load	Error
0x1813	Overcurrent at C/Q (if digital output) – check load	Error
0x1814 to 0x1EFF	Reserved	
0x1F00 to 0x1FFF	Vendor specific	

³⁾ Source: IO-Link Interface Specification V1.1.3, June 2019, Table D.2 - EventCodes for Ports

EventCode ID	Definition and recommended maintenance action	Type
0x2000 to 0x2FFF	Safety extensions	
0x3000 to 0x3FFF	Wireless extensions	
0x4000 to 0x5FFF	Reserved	
0x6000	Invalid cycle time Trigger: SM_PortMode (CYCTIME_FAULT)	Error
0x6001	Revision fault – incompatible protocol version Trigger: SM_PortMode (REVISION_FAULT)	Error
0x6002	ISDU batch failed – parameter inconsistency?	Error
0x6003 to 0xFF20	Reserved	
0xFF21 ¹⁾	DL: Device plugged in ("NEW_SLAVE") – PD stop Trigger: SM_PortMode (COMREADY); see Figure 71 (T10)	Notification
0xFF22 ¹⁾	Device communication lost ("DEV_COM_LOST") Trigger: see Figure 101 (T9)	Notification
0xFF23 ¹⁾	Data Storage identification mismatch ("DS_IDENT_MISMATCH") Trigger: see Figure 104 (T15)	Notification
0xFF24 ¹⁾	Data Storage buffer overflow ("DS_BUFFER_OVERFLOW") Trigger: see Figure 104 (T17)	Notification
0xFF25 ¹⁾	Data Storage parameter access denied ("DS_ACCESS_DENIED") Trigger: see Figure 104 (T29), Figure 105 (T32), Figure 107 (T39)	Notification
0xFF26	Port status changed – Use "SMI_PortStatus" service for port status in detail Trigger: see Figure 101 (T12)	Notification
0xFF27	Data Storage upload completed and new data object available Trigger: see Figure 104 (T26)	Notification
0xFF28 to 0xFF30	Reserved	
0xFF31 ¹⁾	DL: Incorrect Event signalling ("EVENT") Trigger: none	Notification
0xFF32 to 0xFFFF	Reserved	

1) No more required due to SMI Event concept. Not recommended for new implementations.

Example of common port-specific Event Code

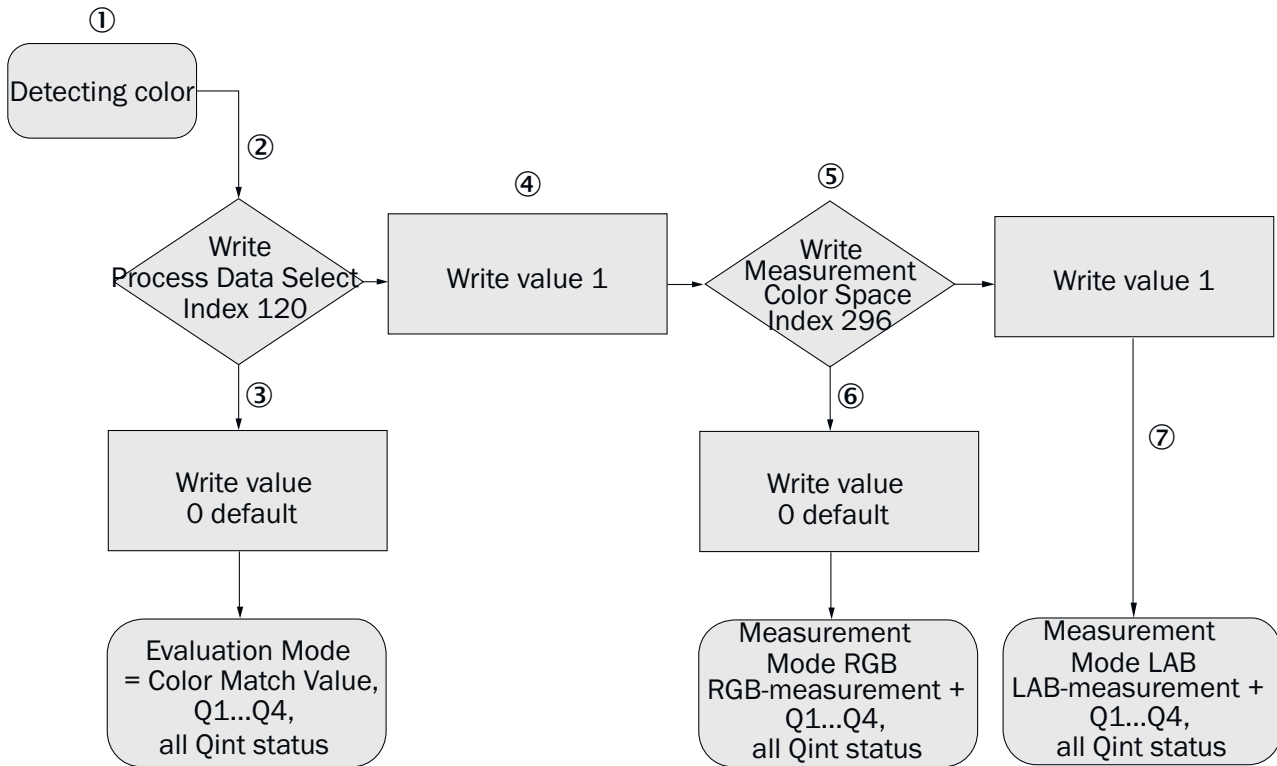
SIG350 sends the port event of a disappearing 'No Device (communication)' error:

▶ LastEvent_Type	'Error'	{...}		STRING
▶ LastEvent_Source	'Master'	{...}		STRING
▶ LastEvent_Qualifier	16#bc		Hex	SINT
▶ LastEvent_Mode	'Event disappears'	{...}		STRING
▶ LastEvent_Instance	'Application'	{...}		STRING
▶ LastEvent_Code	16#1800		Hex	INT

Figure 8: Source: Logix Designer, Studio 5000 using SIG350 as IO-Link Master

12 Use case: Setting the process data

Configuring the process data setting



- ① Align sensor with color object
- ② Set the desired measuring function via **Process Data Select** Index 120
- ③ Write value 0 = **Evaluation Mode** (factory setting)
Transmission of the CMV values (color match) for 4 taught-in colors, as well as the switching states for 24 colors and the diagnostic status of the sensor
- ④ Write value 1 = **Measurement Mode**
Transmission of the color measurement of the current color and its CMV value, as well as the switching states for 24 colors and the diagnostic status of the sensor
- ⑤ Set the desired type of color measurement via **Measurement Color Space** Index 296
- ⑥ Write value 0 = **RGB** (factory setting)
Transfer of the RGB values, i.e. the red, green and blue components of the current color
- ⑦ Write value 1 = **LAB**
Transmission of the Lab color space of the current color and its CMV value, as well as the switching states for 24 colors and the diagnostic status of the sensor

13 Technical data

Table 63: Mechanics/Electronics

Cable length of IO-Link master and IO-Link device	max. 20 m
IO-Link specification	V1.1

14 List of abbreviations

Table 64: List of abbreviations

IODD	IO Device Description	Device description file of an IO-Link device
ISDU	Indexed Service Data Unit	Service data object in IO-Link
COM1 COM2 COM3	SDCI communication mode	COM1 = 4.8 kbit/s COM2 = 38.4 kbit/s COM3 = 230.4 kbit/s
SDCI	Single-drop digital interface	Official (specification) name for IO-Link technology
SDD	SOPAS ET Device Description	Device description file / driver for SICK SOPAS ET software

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I

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